

lsf

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Attribution

lsf is designed for extracting hydrological loading displacement from GNSS vertical position time series analysis based on the least squares fitting method. It is a modified version of Tsview software (http://www-gpsg.mit.edu/~tah/GGMatlab/#_tsview) (Herring, 2003)) and is suitable for batch processing of GNSS time series.

Preparing some relevant files

When implementing this software, readers need to prepare some relevant files:

1. GNSS data file (.pos). *The GNSS position time series at each station are saved in a single file (.pos) and all of them are placed in the folder (pbo/).*
2. Offset data file ('_break.neu'). *The offset data files are saved in '_break.neu' for each station and readers need to manually make these files. Different formats indicate different types of breakpoints, including instrumental offsets (e.g., 2011.37808219178 9999.9999999999 0 0), seismic offsets (e.g., 2011.37808219178 9999.9999999999 0 1), rate changes (e.g., 2011.37808219178 9999.9999999999 1 0), exponential postseismic term (e.g., 2011.37808219178 60 2 0) and logarithmic postseismic term (e.g., 2011.37808219178 60 2 1), note that the number 60 is relaxation time for postseismic deformation.*
3. NTAL (.ntal), NTOL (.ntol) and GIA (*.gia) files. *obtain non-tidal atmospheric and oceanic loading displacements from the German Center for Geoscience (<http://esmdata.gfz-potsdam.de:8080/repository>) and GIA effects from the webpage (<https://www.atmosp.physic.s.utoronto.ca/~peltier/data.php>).*

To acknowledge use of this software,
please cite some of following publications

Herring, T. (2003). MATLAB Tools for viewing GPS velocities and time series. *GPS Solutions*, 7, 194–199. doi: 10.1007/s10291-003-0068-0

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